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Integrated Single-Cell Multi-Omics Analysis Reveals That a CD8⁺ TPex–Monocyte Interaction Axis Coordinates Immune Infiltration in Alzheimer’s Disease

Yusen Zhao, Xinrong Li, Wenbo Dong, Hongbo Zhu, Shuangshuang Wang, Manyi Xu, Yongle Xu, Mengmeng Liu, Junjie Duan, Yujie Liu, Wei Feng, Shangwei Ning *  and Hui Zhi * 

College of Bioinformatics Science and Technology, Harbin Medical University, 157th Rd of Baojian Nangang District, Harbin 150081, China

* Correspondence: ningsw@ems.hrbmu.edu.cn (S.N.); zhihui@ems.hrbmu.edu.cn (H.Z.)

Abstract

Alzheimer’s disease (AD) is a major public health issue, and the role of peripheral immunity in its pathogenesis remains poorly understood. This study conducted a comprehensive re-analysis of publicly available single-cell transcriptomic and chromatin accessibility datasets to investigate immune cell dynamics in AD. By integrating data from cerebrospinal fluid and peripheral blood samples, we constructed a cross-tissue immune cell atlas. Based on Monocle3 pseudotemporal trajectory analysis, we propose the hypothesis that CD8⁺ TEMRA cells in the cerebrospinal fluid may originate from blood-derived CD8⁺ TPex cells. Furthermore, cell–cell communication analysis revealed a potential interaction mechanism whereby CD8⁺ TPex cells secrete MIF signals to activate monocytes, prompting them to release increased levels of inflammatory factors (*IL1B*) and adhesion molecules (*ICAM1*). These inflammatory factors collectively contribute to the disruption of the blood–brain barrier, thereby facilitating immune cell infiltration. Our reanalysis provides a novel interpretation of existing data, establishes a regulatory framework for understanding immune infiltration in AD.

Keywords: Alzheimer’s disease; single-cell multi-omics; immune infiltration

1. Introduction

Alzheimer’s disease (AD), the leading cause of dementia in the elderly population, has emerged as a significant global public health issue. With the ongoing acceleration of global population ageing, the incidence of AD is increasing markedly, resulting in considerable physical, mental, and economic burdens for patients and their families. This rise also imposes substantial challenges to delivering social medical resources. The pathogenesis of AD is characterized by an intricate interplay between genetic and environmental factors. Numerous studies have established a robust association between various genes and the risk of developing AD. Specifically, early-onset familial AD is predominantly linked to mutations in genes such as amyloid precursor protein (*APP*), *PSEN1*, and *PSEN2*, whereas the *APOE* gene plays a crucial role in the manifestation and progression of late-onset sporadic AD. Furthermore, environmental variables, including dietary habits, physical activity levels, and exposure to toxic substances, significantly influence both the risk of developing the disease and the rate of its progression. Despite extensive research endeavors, the precise molecular mechanisms involved in the onset and progression of AD remain inadequately understood.



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In recent years, the blood–brain barrier (BBB) has garnered increasing attention in research concerning AD. As a highly selective semipermeable membrane, the BBB is crucial to maintaining the stability of the internal environment of the central nervous system (CNS) because it effectively prevents harmful substances in the bloodstream from entering the brain. Significant evidence indicates that the BBB in patients with AD causes considerable structural and functional damage, which may facilitate the infiltration of peripheral immune cells into the brain [1]. Historically, it was believed that under normal physiological conditions, immune cells were unable to penetrate the CNS. However, emerging studies have revealed that in the context of neurological disorders such as AD, knockout of the *CXCR3* gene in blood cells can markedly ameliorate memory impairment [2]. This finding suggests the potential for immune cells to traverse the BBB and enter the brain. These infiltrating immune cells may contribute to neuroinflammation, a characteristic pathology of AD. Within the pathogenesis of AD, immune cells function as a double-edged sword: on the one hand, they can initiate an immune response to clear amyloid plaques and damaged neurons; on the other hand, overactivation of the immune system may lead to excessive inflammation, resulting in neuron damage.

The advancement of single-cell multi-omics technologies has provided new perspectives for deciphering the immune mechanisms of Alzheimer’s disease (AD). For instance, Ramakrishnan et al. [3] employed single-cell ATAC-seq and RNA-seq to systematically characterize the epigenetic landscape of peripheral immune cells in AD, revealing widespread chromatin accessible regions in AD patients, particularly in CD8⁺ T cells and monocytes (Ramakrishnan et al., 2024). However, how these epigenetic alterations specifically drive functional changes in immune cell subsets, enabling them to cross the blood–brain barrier and dynamically infiltrate the central nervous system, remains unclear. We performed an integrated reanalysis of existing, publicly available single-cell omics datasets from AD patients and controls to construct an immune cell atlas of cerebrospinal fluid (CSF) and peripheral blood in AD. We focus on elucidating the migratory trajectories, cellular interactions, and molecular regulatory networks of CD8⁺ T-cell and monocyte subsets in AD. Our work aims to uncover the detailed mechanisms of immune infiltration from a dynamic and functional perspective.

2. Results

2.1. *scRNA-Seq Reveals Immune Cell Heterogeneity and Dynamic Changes in the Blood and Cerebrospinal Fluids of AD Patients*

After data quality control, we retained 17,209 high-quality single cells from the CSF samples. Unsupervised clustering and annotation based on established marker genes delineated eight major immune cell types in the CSF compartment, encompassing key subsets of both the adaptive and innate immune systems (Figure 1A). The specific marker genes used for annotation were: *CD3D* and *CD4* for CD4⁺ T cells; *CD3D*, *CD8A* and *CD8B* for CD8⁺ T-cell subsets (including effector memory T cells, TEMRA); *NCAM1* (*CD56*) for natural killer (NK) cells; *CD14* and *FCGR3A* (*CD16*) for monocyte subsets; *CD11C* (*ITGAX*) and *IL3RA* for dendritic cells (DCs); *CD19*, *CD79A*, *CD79B* and *MS4A1* for B cells; and *MZB1* and *JCHAIN* for plasma cells. In the CSF of the AD group, we observed a marked increase in the abundance of terminally differentiated CD8⁺ TEMRA cells ($p = 0.039$) (Figure 1B). Differential gene expression analysis revealed that CD8⁺ TEMRA cells specifically express the inflammatory gene *DUSP2* and the cytotoxic gene *GZMK* (Supplementary Figure S2). This cellular shift may be linked to AD-associated neuroinflammation and memory impairment.

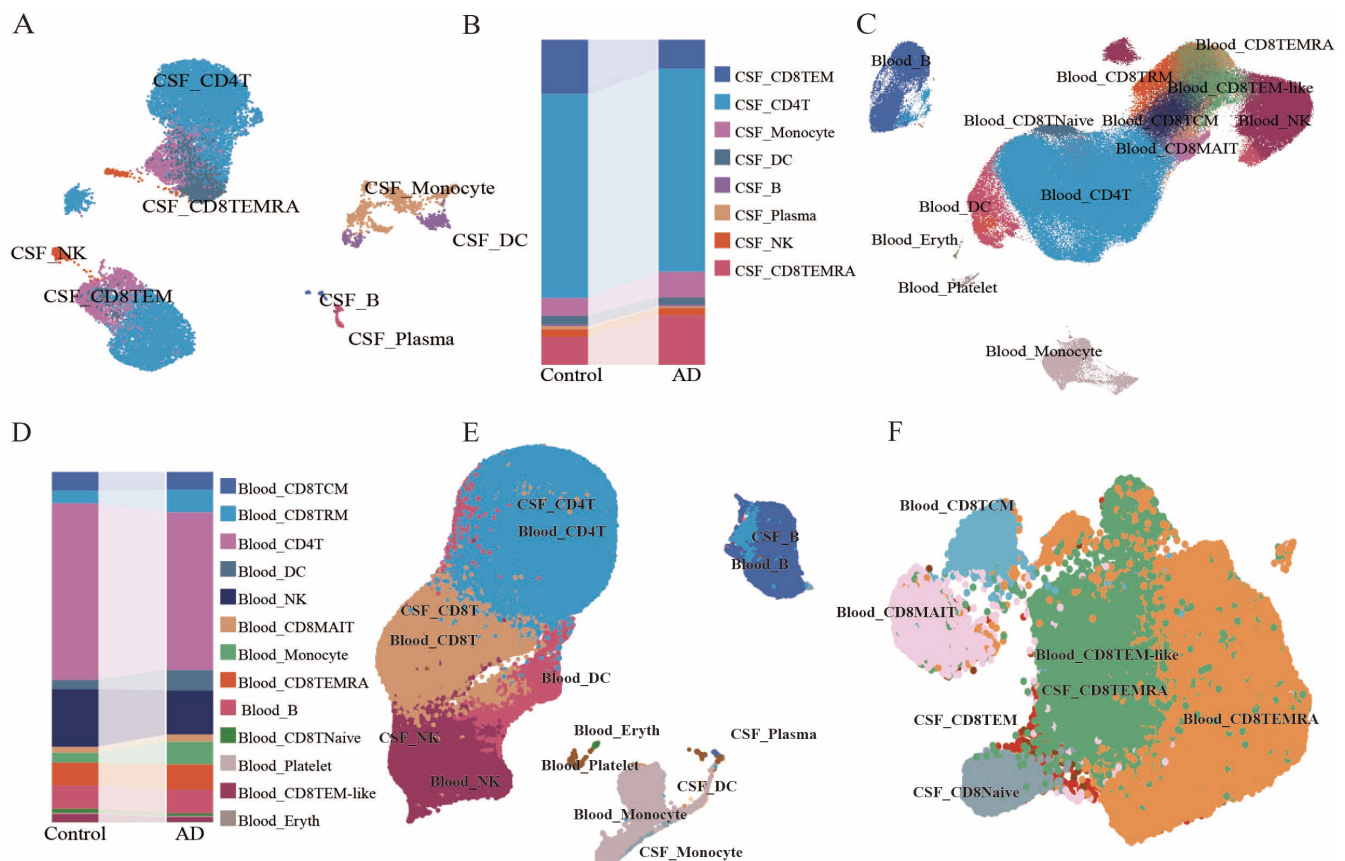


Figure 1. Single-cell transcriptome atlas of AD cerebrospinal fluid and blood. (A) UMAP plot of the single-cell transcriptome of AD cerebrospinal fluid. (B) Sankey diagram of the relative abundance of cell subsets in AD cerebrospinal fluid. (C) UMAP plot of the single-cell transcriptome of AD blood. (D) Sankey diagram of the relative abundance of cell subsets in AD blood. (E) UMAP of single-cell transcriptome integration analysis of AD blood and cerebrospinal fluid. (F) UMAP of single-cell transcriptome integration analysis of CD8T cells in AD blood and cerebrospinal fluid.

To explore potential connections between central and peripheral immunity, we extended the analysis to peripheral blood samples. For the blood samples, 270,883 cells were detected after rigorous quality control and were partitioned into 13 cell types. Most of these types overlapped with those identified in the CSF, with additional populations unique to the blood, including CD8⁺ mucosal-associated invariant T (MAIT) cells (marked by *SLC4A10*, *KLRB1*, *RORC* and *RORA*), platelets (expressing *CD41*, *CD61* and *CD62P*) and erythrocytes (expressing *CD36*, *HBA1* and *HBA2*) (Figure 1C). Notably, the CD8TEM-like subset in the peripheral blood showed specific enrichment in AD patients, characterized by high *GZMK* expression and functional association with T-cell differentiation (Supplementary Figure S3). This disease-specific subset was consistently identified after rigorous quality control and batch correction, supporting its biological relevance in AD pathogenesis. The were changes in abundance of specific subsets in the blood, such as CD8⁺ TEM-like cells ($p = 0.015$) (Figure 1D). Integrated analysis of the blood and cerebrospinal fluid transcriptomes revealed that CD8⁺ TEM-like cells in the blood overlapped with CD8⁺ TEMRA cells in the cerebrospinal fluid within the same UMAP space (Figure 1E,F), suggesting potential functional interactions and directional migration between the peripheral and central immune systems. The findings of this study provide a new research perspective and an important theoretical basis for further exploration of the pathogenesis of AD and identification of potential therapeutic targets.

2.2. Cell Trajectory and Subpopulation Analysis Suggests a Potential Origin of Cerebrospinal Fluid CD8⁺ TEMRA from Blood CD8⁺ TEM-like in AD

To further explore the potential connection between CD8⁺ TEM-like cells in the blood and CD8⁺ TEMRA cells in the cerebrospinal fluid, we performed a reclustering analysis on these two cell populations. The reclustering of CSF CD8⁺ TEMRA cells identified two subpopulations (Figure 2A): CD8⁺ TEMRA cluster 1 highly expressed genes related to enhanced immune activity (*CCL3L1*, *CD8A*, and *ERAP2*) [4,5], with functional enrichment in the amphisome membrane and MHC class I protein binding pathways (Supplementary Figure S4A). The abundance of CD8⁺ TEMRA cluster 2 increased significantly, characterized by high expression of virus-related genes (*NKG7*, *GZMA*, and *GZMH*) and oxidative stress genes (*GSTM1*), with functional enrichment in the cell killing, cytosolic ribosome and oxidative phosphorylation pathways (Figure 2B; Supplementary Figure S4B) [6,7]. The reclustering of blood CD8⁺ TEM-like cells also identified two subpopulations (Figure 2C): The control group had fewer cells, mainly in cluster 1, and the functions were significantly enriched in the plasma membrane signaling receptor complex and paranode region of axons. In the AD group, the abundance of CD8⁺ TEM-like cluster 1 cells decreased significantly (high expression of *CD8A*, *CD8B*, *CXCR4* chemokines, and the *JUN* proliferating gene), whereas that of other subpopulations increased, with functional enrichment in T-cell differentiation (Supplementary Figure S4C). CD8TEM-like cluster 2 highly expressed cytotoxic-related genes (*GZMA* and *GZMB*), *CXCR3* chemokines, with functional enrichment in aerobic respiration and ATP biosynthetic process, maintaining cell survival and the expression of toxic factors (Figure 2D; Supplementary Figure S4D) [8–10].

For further in-depth research, we integrated the single-cell transcriptome data from CSF and blood and performed Monocle3 (v1.3.7)-based cell trajectory inference analysis [11]. Trajectory analysis suggested a potential developmental link, proposing that cerebrospinal fluid CD8⁺ TEMRA cells may originate from blood CD8⁺ TEM-like cells. Blood CD8⁺ TEM-like cluster 1 may damage the blood–brain barrier (BBB) via inflammatory factors and then enter the CSF and differentiate into CD8⁺ TEMRA cells; blood CD8⁺ TEM-like cluster 2 can also cross the BBB into brain tissue (Figure 2E). Genes in the cell trajectory were divided into two categories: functional enrichment analysis revealed one category related to “immune regulation” and “leukocyte migration” and the other to “leukocyte differentiation and toxicity” (Figure 2F–H). This discovery provides key clues for a deep understanding of the cross-tissue dynamic changes and functional regulation of immune cells in the pathogenesis of AD and is expected to lay an important foundation for the development of subsequent targeted intervention strategies.

2.3. CD8⁺ TPex Cells Mediate Neurodegenerative Diseases by Regulating Blood–Brain Barrier Infiltration and Immune Function

Cell trajectory analysis identified blood CD8⁺ TEM-like as a key subset crossing the BBB. Using the hdWGCNA R package (v0.4.04) [12] with genes detected in at least 5% of the cells as input, we constructed a hierarchical clustering tree (threshold = 2) and identified eight gene co-expression modules (Figure 3A). The turquoise, blue, brown, and black modules were specifically expressed in CD8⁺ TEM-like cluster 1 (Figure 3B). We calculated module eigengene (ME) values to screen characteristic genes and combined them with protein–protein interaction (PPI) analysis to define seven hub genes *CCR7*, *CD69*, *KLRB1*, *KLRF1*, *CXCR4*, *RGS1*, and *GZMK*, which are highly specific to cluster 1 and have high chromatin accessibility near promoters (Figures 3C and 4A). Chromatin transcription factor enrichment identified *TCF7*, *ZEB1*, and *RUNX1* as the top regulators; *TCF7* (highly expressed in the transcriptome) maintains cell stemness (Figure 4B). Exhaustion scoring showed that CD8⁺ TEM-like cluster 1 had higher exhaustion scores than cluster 2 (higher toxicity; Figure 3D). Cells highly expressing the *TCF7*, *CCR7*, and exhaustion genes were

defined as CD8⁺ TPex cells [13–17], with the expression of the *CCR7/TCF7* hub genes being downregulated in trajectories, suggesting differentiation into terminal cells (Figure 3C).

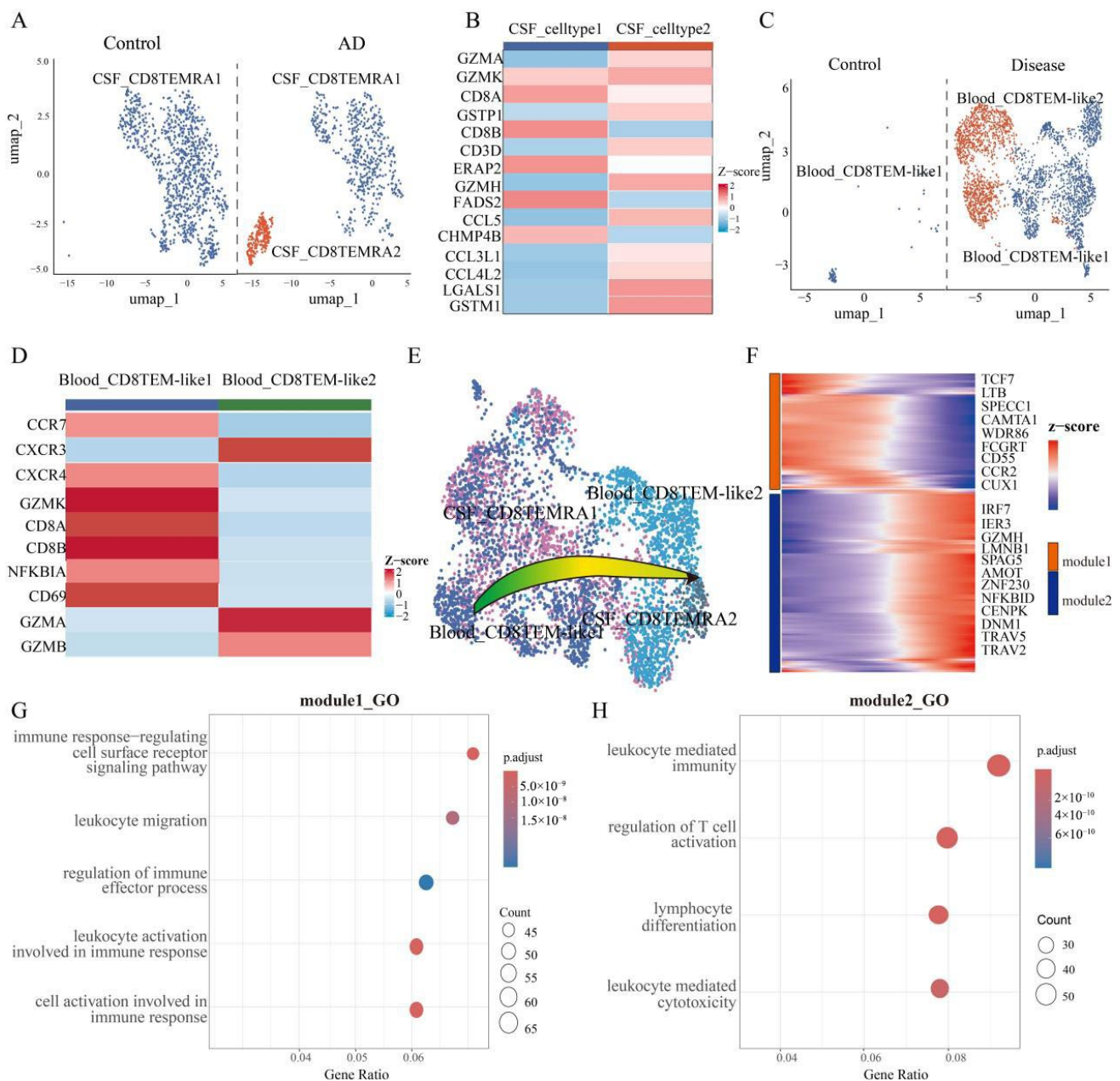


Figure 2. Single-cell transcriptome atlas of CD8TEMRA in the cerebrospinal fluid and CD8TEM in the blood. (A) UMAP plots of the single-cell transcriptome of CD8TEMRA in the cerebrospinal fluid of AD patients, including the control group and the disease group. (B) Specific markers of CD8TEMRA cell subsets in the cerebrospinal fluid of AD patients. (C) UMAP plots of the single-cell transcriptome of CD8TEM-like in the blood of AD patients, including the control group and the disease group. (D) Specific markers of CD8TEM-like cell subsets in the blood of AD patients. (E) Schematic diagram of the results of the pseudotemporal order analysis after integrating the single-cell transcriptome data of CD8TEMRA in the cerebrospinal fluid and CD8TEM-like in the blood. (F) Heatmap of gene distribution in the trajectory after integration of blood CD8TEM-like and cerebrospinal fluid CD8TEMRA cells. (G) After the integration of blood CD8TEM-like and cerebrospinal fluid CD8TEMRA cells, the genes were classified in the trajectories of module1 functional enrichment analysis dot plots. (H) After the integration of blood CD8TEM-like and cerebrospinal fluid CD8TEMRA cells, the genes were classified in the trajectories of module2 functional enrichment analysis dot plots.

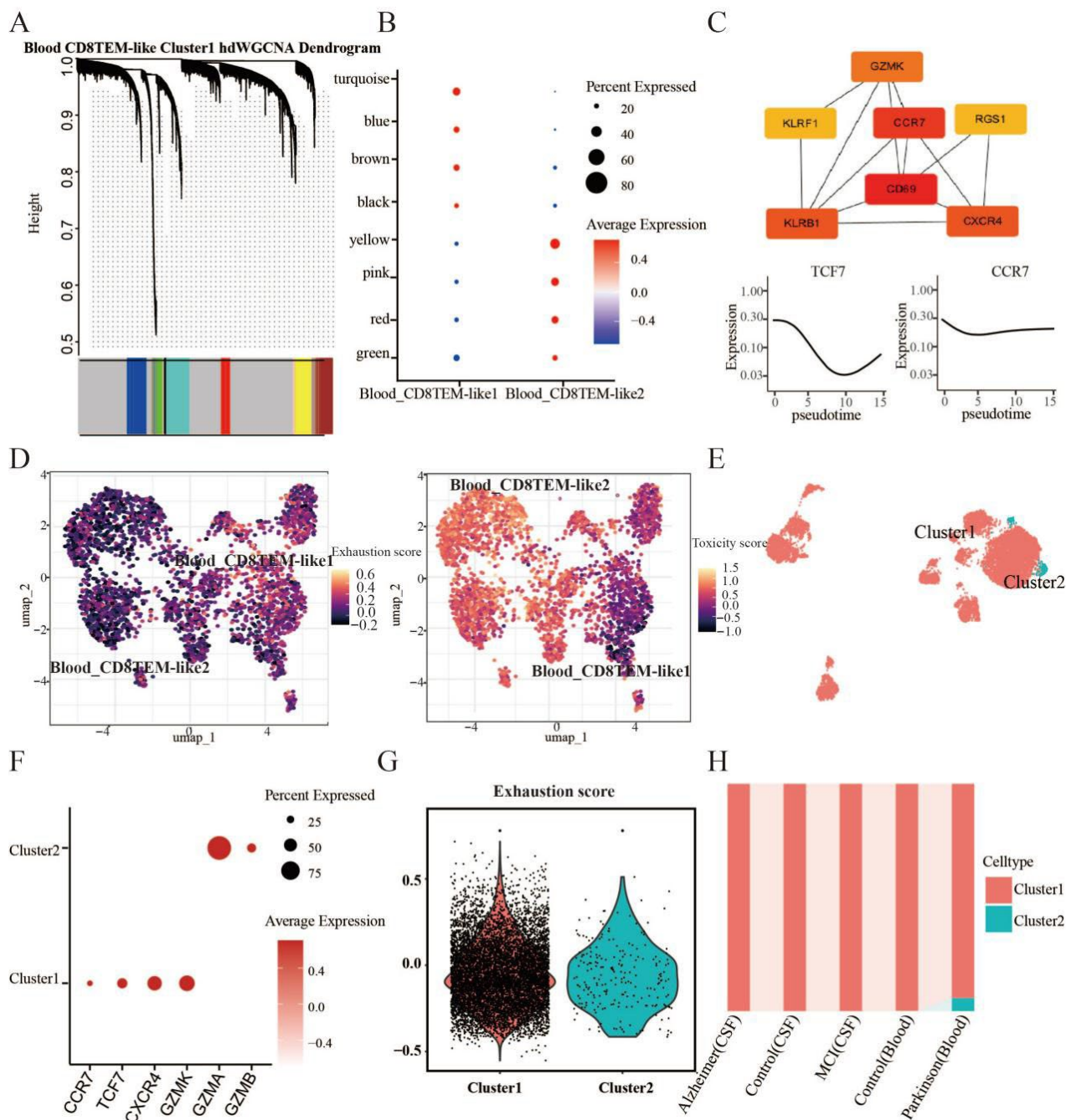


Figure 3. Analysis of gene co-expression networks (hdWGCNA) in blood single-cell transcriptomes and transcriptome validation of Parkinson's disease. (A) Gene co-expression matrix (with horizontal axis representing genes and vertical axis representing correlations). (B) Dot plot of the specific expression of the gene co-expression module in the CD8TEM-like subpopulation. (C) Hub genes obtained by hdWGCNA for CD8TEM-like cluster 1 subset in single-cell transcriptome of blood (top); the continuous expression of hub genes in the cell trajectory (bottom). (D) Exhaustion scores of each subgroup of CD8TEM-like subpopulation (left); toxicity scores of each subgroup of CD8TEM-like subpopulation (right). (E) UMAP map of the reclustering analysis of transitional CD8T cells in Parkinson's disease (Qinghua Jiang et al. [18]). (F) Dot plot of hub genes in the CD8TEM-like cluster 1 gene subgroup in transitional CD8T-cell subsets of Parkinson's disease. (G) The exhaustion score of transitional CD8T-cell subsets in Parkinson's disease. (H) The transitional CD8T-cell subset of Parkinson's disease is a cellular component in the blood (healthy control group and disease group) and cerebrospinal fluid (healthy control group, MCI disease group and AD group).

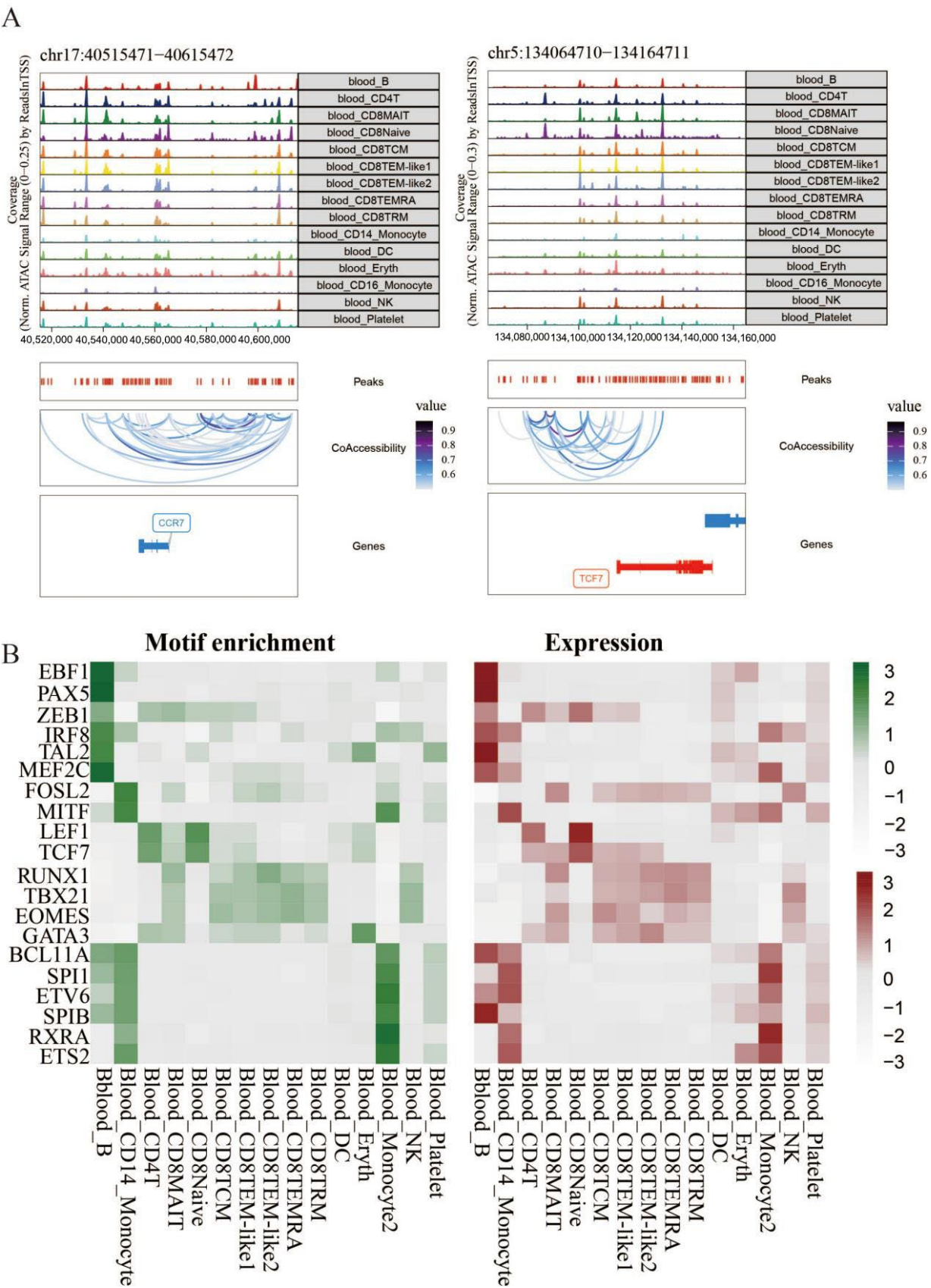


Figure 4. Analysis of single-cell chromatin accessibility in AD blood. (A) Peak co-expression map of hub genes in CD8TEM-like cluster 1 in AD blood in scATAC-seq analysis. (B) Heatmap representing the enrichment of selected TF motifs in 3 sets of peaks (left), as well as expression in different subtypes (right).

Cross-disease comparison: PD studies have shown that memory-related cerebrospinal fluid CD8⁺ TEMRA (David Gate et al.) and blood GZMK⁺ transitional CD8⁺ T cells (Qinghua Jiang et al.) [18] are clonal and migratory, which is consistent with the CD8⁺ TPex cells in this study having a marker (GZMK). The reclustering of PD transitional CD8⁺ T cells identified subpopulations with high expression and exhaustion scores for CD8⁺ TPex markers, confirming the presence of CD8⁺ TPex in both AD and PD patients (Figure 3E–H). The functional enrichment of CD8⁺ TPex was linked to T-cell differentiation, NF-kappaB inhibition, and stem cell proliferation (Supplementary Figure S4C).

We further analyzed brain vascular endothelial cells via hdWGCNA: a total of 9 modules were identified, and 10 hub genes were screened after LASSO regression/PPI. *TAGLN* and *FLNA* (which enhance cell adhesion via integrin/membrane protein binding) [19,20] were enriched in “muscle contraction” and “homotypic cell–cell adhesion” (Supplementary Figure S5A–C). The results indicate that the identity of vascular endothelial cells was lost and their functions were disrupted in AD. Given that AD-related endothelial TDP-43 depletion activates NF-kB to disrupt the BBB [21], we propose the following model: endothelial cell dysfunction in AD leads to the destruction of the blood–brain barrier; with the assistance of CXCR4, CD8⁺ TPex infiltrates the CSF, differentiates into CD8⁺ TEMRA, and impairs neurons.

2.4. Inflammatory Signaling Centered on the MIF Pathway Promotes Blood CD8⁺ TPex Immune Infiltration into Cerebrospinal Fluid in AD

To explore the intercellular interactions driving CD8⁺ TPex infiltration, we used CellChat (v1.6.1) [22] to analyze ligand–receptor networks. The results revealed that blood CD8⁺ TPex cells interact with monocytes, CD8⁺ TEM-like cluster 2, CD8⁺ MAIT, and CD8⁺ TCM cells, primarily via the MIF signaling pathway (Figure 5A). Ligand–receptor enrichment confirmed that CD8⁺ T cells are enriched in the MIF, ANNEXIN, PAR, and IL16 pathways, whereas monocytes are enriched in the MIF, GALECTIN, TNF, and BAFF inflammatory pathways (Figure 5C,D; permutation test, adjusted $p < 0.05$). In the MIF pathway, CD8⁺ TPex acts as a “transmitter” (secreting MIF ligands), a “receiver” (expressing MIF receptors CD74/CD44 and CD74/CXCR4) and an “influencer” (interacting with other cells to promote cell migration), whereas monocytes act mainly as “receivers” and “targets” (Figure 5E). MIF-related genes (MIF, CD74, and CD44) were highly and coordinately expressed across interacting cell subsets (Figure 5F), identifying the MIF signaling axis as a key mediator of potential crosstalk in the AD microenvironment. These interactions synergistically promote infiltration via two mechanisms: (1) MIF signaling upregulates inflammatory factors (e.g., *TNF* and *IL16*) to disrupt BBB integrity; (2) it enhances cell migration to facilitate the CD8⁺ TPex crossing of the BBB into the CSF.

2.5. The MIF Signaling Pathway Promotes Monocyte Migration

Given the central role of MIF signaling in CD8⁺ TPex–monocyte interactions, we next sought to characterize the monocyte subsets involved. Thus we conducted a reclustering analysis of monocyte cells, and the results showed that the two cell subpopulations were identified as CD14 monocytes and atypical CD16 monocytes, respectively. In the disease group, we found that the proportion of atypical CD16 monocytes decreased and that of CD14 monocytes increased, enhancing the phagocytic ability of cells and thereby eliminating damaged and apoptotic cells (Figure 6A,B).

Through differential analysis, we found that CD14 monocytes highly express a series of chemokines, such as *CCL4*. These chemokines can enable monocytes to quickly reach the inflammatory site and promptly eliminate pathogens. Meanwhile, monocyte CD14 is highly expressed in monocytes to promote inflammation and cell adhesion and to inhibit the angiogenesis genes, *THBS1* and *FLT1* (VEGF receptor). The *FLT1* gene can capture free

VEGFA, thereby inhibiting the binding of *VEGFA* genes to *VEGFR2* receptors in vascular endothelial cells and exacerbating angiogenesis disorders. This leads to the destruction of the blood–brain barrier structure [23]. In addition, the expression levels of inflammation-related genes in CD14 monocyte cells, such as *IRAK2* and *EBI3*, also increased significantly. Atypical CD16 monocytes, with high expression of stress response genes, *DDIT4* (regulated by hypoxia or oxidative stress) [24], and genes involved in cell adhesion (*JUP*), indicates the presence of a hypoxic environment during disease occurrence, leading to functional changes in atypical CD16 monocytes and a decline in their phagocytic capacity (Figure 6C,E).

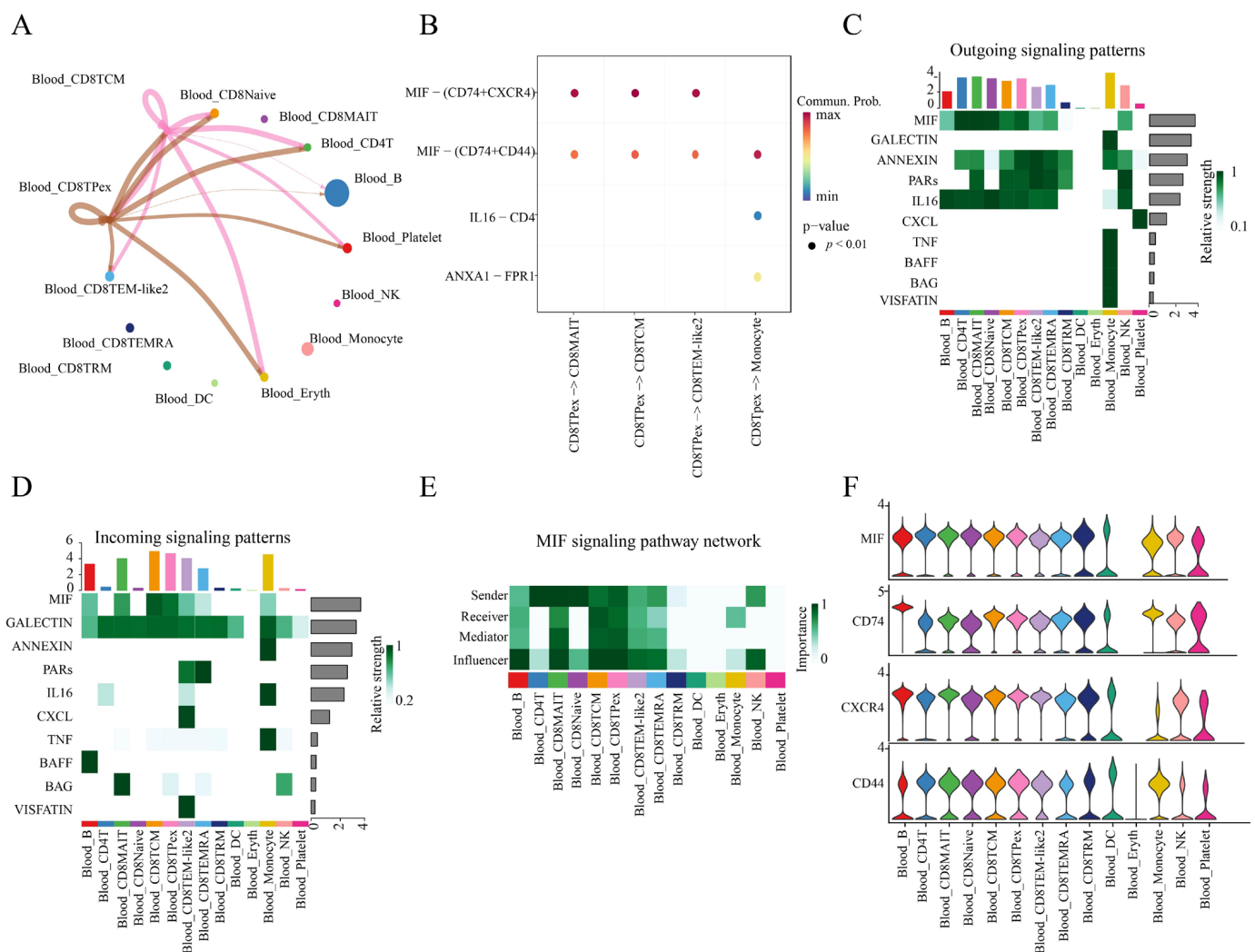


Figure 5. Interaction analysis of various cell subsets in the single-cell transcriptome of blood. (A) Interaction analysis between CD8TPex in the blood of AD patients and other cells (the thicker the brown line, the stronger the cell interaction). (B) Enrichment dot plot of ligand–receptor pairs for the interaction between CD8TPex and other cells in the blood of AD patients. (C) Heatmap of output signals of signaling pathways in which various cell subsets in the blood of AD patients participate (represented by the bar chart above). (D) Heatmap of input signals of signaling pathways in which various cell subsets in the blood of AD patients participate. (E) The importance of various cell subsets in the blood of AD patients in the four different roles of the MIF signaling pathway (sender; receiver; mediator; influencer). (F) Gene expression levels related to the MIF signaling pathway in various cell subsets in AD.

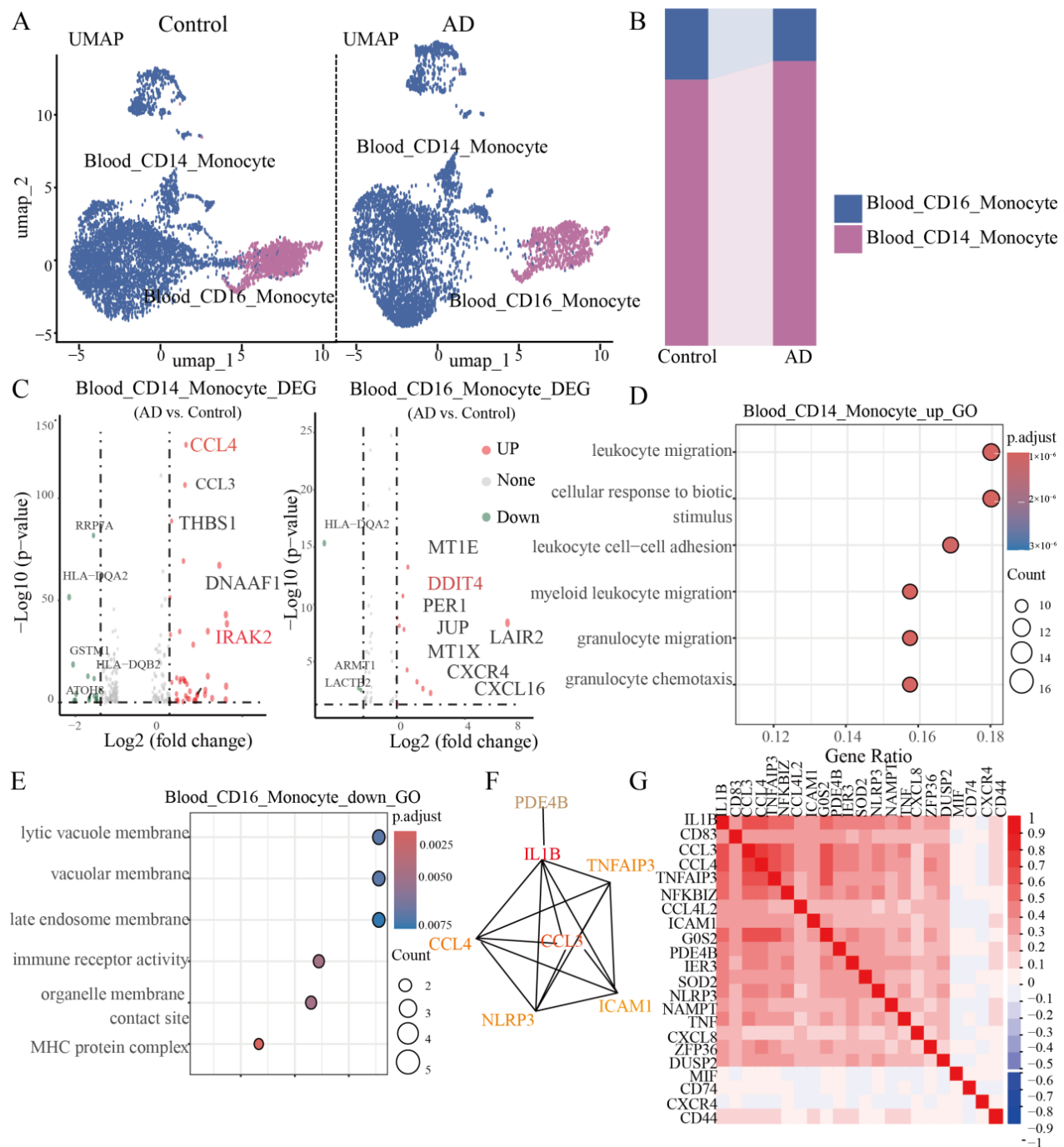


Figure 6. Single-cell transcriptome analysis of monocyte cells in blood of AD patients. **(A)** UMAP plot of the reclustering analysis of monocyte cells in the blood. **(B)** Sankey diagram of the composition analysis of monocyte cells in the blood. **(C)** Volcano plot of the differential gene analysis of each subpopulation of monocyte cells in the blood. **(D)** Result diagram of the functional enrichment analysis in the CD14 monocyte subpopulation in the blood. **(E)** Result diagram of the functional enrichment analysis in the CD16 monocyte subpopulation in the blood. **(F)** Co-expression network diagram of hub genes in the CD14 monocyte subpopulation in the blood. **(G)** Heatmap of the correlation analysis between genes related to the MIF signaling pathway and hub genes of monocyte cells.

The possible associations among genes in monocyte cells were investigated through gene co-expression network analysis (hdWGCNA). A total of 12 gene co-expression modules were identified. Among these, modules that were not expressed in CD14 monocyte

cells were excluded. For the remaining modules, the top genes were selected as hub genes. Through PPI network analysis, we subsequently identified seven overlapping genes (IL1B, TNFAIP3, CCL3, CCL4, NLRP3, ICAM1, and PDE4B) that were differentially expressed in CD14 monocyte cells, and their expression levels were significantly upregulated in the disease group (Figure 6F). Further functional enrichment analysis showed that these seven genes were significantly enriched in the GO term of “leukocyte migration”, suggesting that these seven key genes are closely related to the positive regulation of the immune response (Figure 6D).

In this study, we observed that MIF-related genes presented high expression levels in both CD14 monocyte cells and CD16 monocyte cells and that there was a significant correlation between these genes and the hub genes of monocyte cells. This result indicates that communication between cells through the MIF signaling pathway may promote the migration process of monocyte cells (Figure 6G).

2.6. Monocyte Cells in the Blood Enter the Cerebrospinal Fluid to Maintain a Stable Immune Environment

Building upon the functional evidence that blood CD14⁺ monocytes are primed for migration, we next sought to investigate whether these cells undergo actual tissue translocation. Moreover, single-cell transcriptome integration analysis shows that monocytes in the blood and cerebrospinal fluid are adjacent in the same UMAP dimensionality reduction space, indicating that monocyte cells in these two tissues have similar biological functions. We conducted a reclustering analysis of cerebrospinal fluid monocyte cells and identified a total of two cell subpopulations, CD14 monocytes and intermediate monocytes (where monocytes express both the CD14 and CD16 genes). In addition, in the disease group, the proportion of CD14 monocyte cells decreased, whereas the proportion of intermediate monocytes increased, reflecting dynamic changes in the immune microenvironment (Figure 7A–C).

Through functional enrichment analysis of the two types of cells, CD14 monocyte cells were significantly enriched in the term “oxidoreductase activity acting on a heme group of donors”, and the genes related to this GO term were significantly downregulated in the disease group, which may lead to the continuous occurrence of inflammatory responses, a decline in immune function, and neuronal damage. The “innate immune response regulation” and “leukocyte differentiation regulation” of intermediate monocytes were significantly enriched, revealing the association between intermediate monocytes and the enhancement of the immune system (Figure 7D,E).

Functional enrichment results indicated that the two monocyte subtypes in the cerebrospinal fluid (CSF) exhibited opposing functional states. Furthermore, trajectory inference analysis performed after integrating single-cell transcriptomic data from blood and CSF revealed that intermediate monocytes in the CSF likely originate from CD14⁺ monocytes in the blood (Figure 7F). We propose a hypothesis that due to the decline in immune function of monocytes within the CSF, monocytes in the blood interact with CD8⁺ TEM-like cells. This interaction promotes the high expression of inflammatory factors and cell adhesion-related genes, subsequently disrupting the blood–brain barrier. This disruption allows monocytes to migrate into the CSF, where they differentiate into intermediate monocytes and enhance the immune functionality of the CSF. In addition, we also found that intermediate CD16 monocytes in the cerebrospinal fluid highly express the *CXCL16* gene [25], which may interact with the CD8⁺ TEMRA cluster 2 subgroup through the *CXCL16*–*CXCR6* axis, promoting the release of toxic molecules by CD8⁺ TEMRA cells, intensifying neuroinflammation and accelerating disease progression.

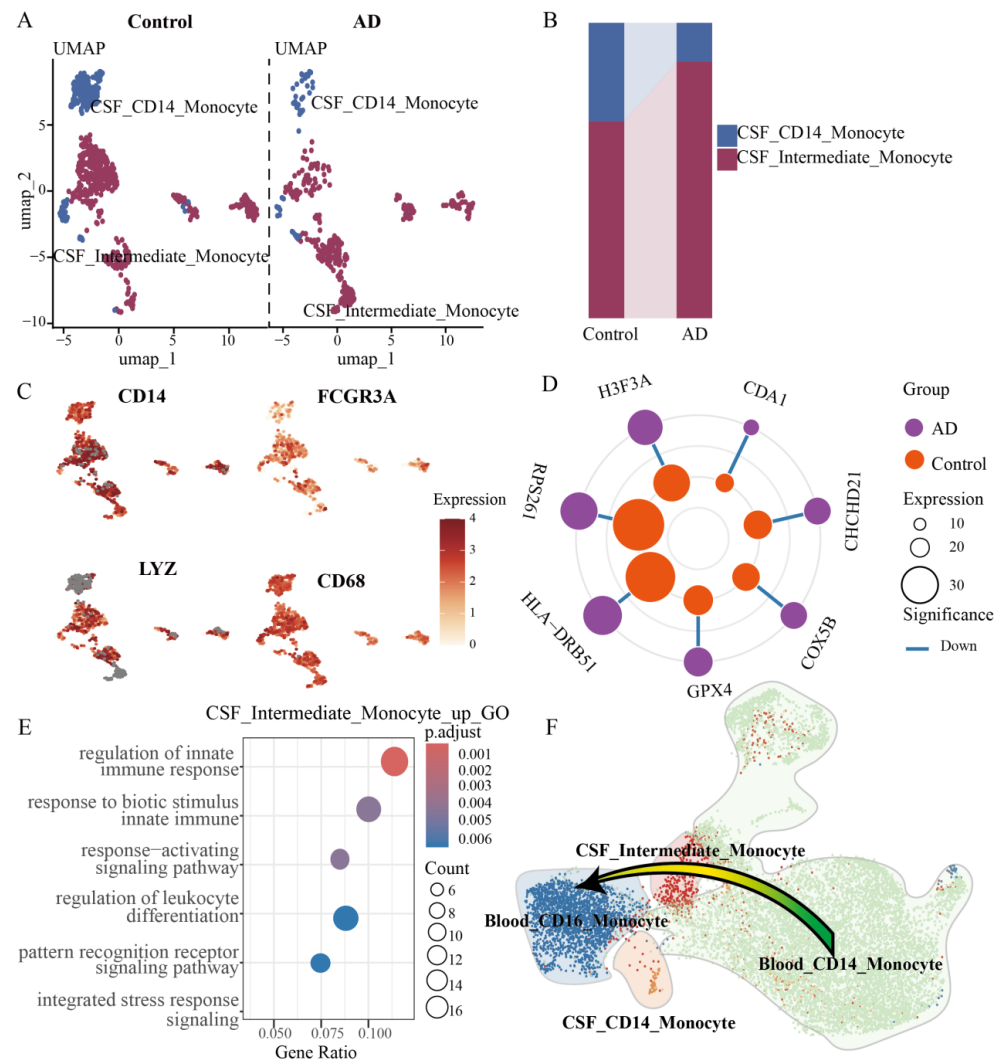


Figure 7. Transcriptome analysis of monocyte cells in cerebrospinal fluid of AD patients. (A) Single-cell transcriptome atlas of monocyte cells in the cerebrospinal fluid. (B) Sankey diagram of the composition analysis of monocyte cells in the cerebrospinal fluid. (C) Feature plot of marker genes of monocyte cells in the cerebrospinal fluid. (D) Circular dumbbell plot of the differential gene analysis of the CD14 monocyte subset in the cerebrospinal fluid (colors represent the control group and the disease group; the size of the circle represents the gene expression level; lines indicate whether the gene is upregulated or downregulated). (E) Functional enrichment dot plot of the median monocyte subset in the cerebrospinal fluid. (F) Schematic diagram of the cell trajectory analysis after integrating monocyte cells in the cerebrospinal fluid and those in the blood.

2.7. scATAC-Seq Analysis Revealed the Regulatory Network of Monocyte Transcription Factors

Motif enrichment analysis of differential chromatin accessibility was conducted for each cell type. The results indicated that CD14 monocytes were specifically enriched with the transcription factors *CEBPB* and *CEBPE*, whereas CD16 monocytes were significantly enriched with the transcription factors *CEBPD* and *CEBPA*. Moreover, AP-1 family transcription factors (*JUN*, *FOS*, *JUNB*, and *JUND*) and NFκB transcription factors were significantly enriched in both CD14 monocyte and CD16 monocyte cells [26–28]. These transcription factors are closely related to the expression of inflammatory genes and cell migration and are consistent with the results of single-cell transcriptome transcription factor analysis. TF footprinting indicates that these transcription factors have high activity in specific cell types (Figure 8A,B). We conducted relevant analyses on key TF–genes. The transcription factors significantly enriched in monocyte cells were mainly related to the reg-

ulation of inflammatory factors, chemokines, and genes related to cell signal transduction. This finding indicates that monocyte cells entering the cerebrospinal fluid maintain the immune microenvironment of brain tissue and the long-term disruption of the blood–brain barrier. This enables more harmful factors to enter the cerebrospinal fluid, exacerbating the development of AD (Figure 8C).

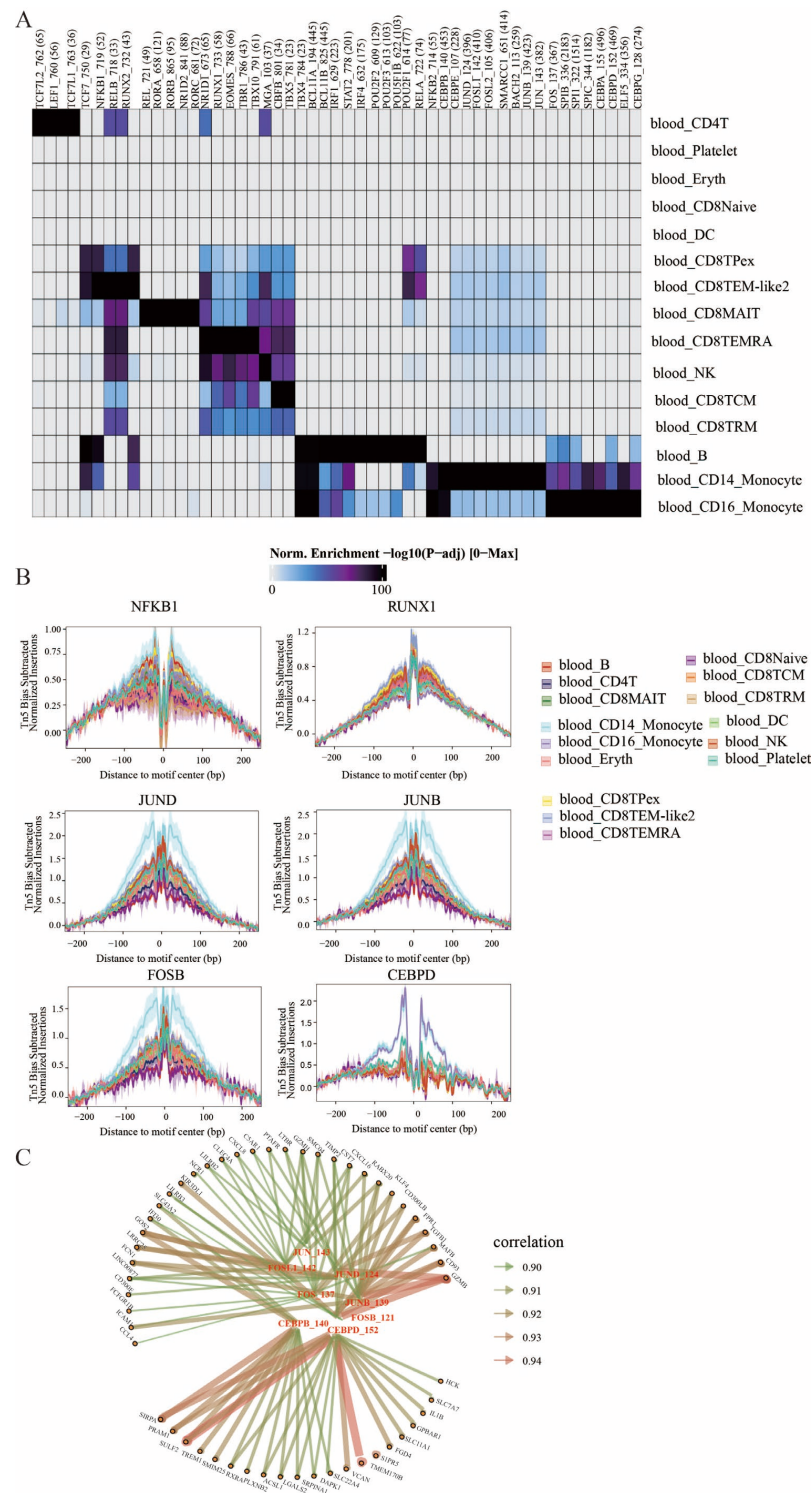


Figure 8. Analysis of chromatin accessibility in the single-cell transcriptome of AD blood. (A) Heatmaps of TF enrichment in scATAC-seq analysis of various subpopulations in AD blood. (B) Tn5 bias-subtracted TF footprinting for hyperactivated TFs in various subtypes. (C) The TF-gene regulatory network of blood monocyte cells in AD.

3. Discussion

This study presents a comprehensive single-cell transcriptomic analysis of peripheral blood and cerebrospinal fluid (CSF) from Alzheimer's disease (AD) patients, proposing a novel model in which CD8⁺ TPex cells interact with monocytes to drive immune infiltration. This integrated analysis provides new insights into the dynamic alterations in peripheral immunity in AD.

This study delineates a multi-step pathogenic trajectory: Blood-derived CD8⁺ TPex cells, characterized by high GZMK expression, upregulate programs associated with signal transduction and inflammation, potentially aided by chemokine signaling and other immune cells. These cells subsequently develop an integrin-mediated adhesive profile, facilitating their migration across the compromised blood–brain barrier (BBB) and infiltration into the CSF. Within the CSF, these cells differentiate into CD8⁺ TEMRA cells, which express high levels of cytotoxic mediators. Given the established association between T-cell cytotoxicity and neuronal injury, this differentiation process is posited to trigger a cascade of events linked to AD pathology, including A β deposition and exacerbated memory impairment. Furthermore, the detection of CD8⁺ TPex cells in Parkinson's disease suggests that this cell type may be associated with memory impairment across a spectrum of neurodegenerative disorders, indicating a potentially shared immunological mechanism.

This study also reveals the phenomenon of monocyte tissue migration accompanying the infiltration of CD8⁺ TPex cells. We hypothesize that monocytes enter the cerebrospinal fluid compartment and differentiate into intermediate monocytes, potentially participating in the pathological process by sustaining immune clearance functions. Comprehensive research data indicate that inhibiting the infiltration of CD8⁺ TPex cells may serve as a potential therapeutic target for alleviating memory impairment in Alzheimer's disease patients. Through single-cell chromatin accessibility analysis, we further identified cell type-specific transcription factors: CD8⁺ TPex cells were significantly enriched for transcription factors associated with cellular stemness (*TCF7*, *RUNX1*, and *ZEB1*), whereas monocytes primarily showed enrichment in transcription factors related to cell proliferation (*JUN* and *FOS*) and proinflammatory responses (*CEBPD*, *CEBPB*, and *NFKB*). The activation states of these transcription factors are closely linked to cellular activity and migratory capacity. Their identification not only elucidates the potential molecular mechanisms underlying immune cell infiltration in Alzheimer's disease but also provides new directions for future targeted therapeutic interventions.

While our study provides a dynamic model of immune cell infiltration in Alzheimer's disease through integrative multi-omics analysis, several limitations should be acknowledged. Firstly, the pseudotemporal trajectory inference, while suggestive, does not conclusively prove a direct lineage relationship between blood CD8⁺ TPex cells and cerebrospinal fluid CD8⁺ TEMRA cells, which is constrained by the lack of TCR sequencing data from matched blood and cerebrospinal fluid samples of the same individuals. Secondly, our reanalysis was limited by the availability of public datasets. The absence of matched single-cell chromatin accessibility data from the cerebrospinal fluid of Alzheimer's disease patients hindered a more comprehensive epigenetic comparison between peripheral and central immune cells.

In future research, we will collect paired blood and cerebrospinal fluid samples from the same patients for single-cell TCR (scTCR) sequencing to validate the migration trajectories and clonal relationships proposed in our model. Additionally, we will isolate brain tissues from Alzheimer's disease patients and perform single-cell spatial transcriptomics to verify the interactions between CD8⁺ TPex cells and monocytes near the blood–brain

barrier. Overcoming these data limitations is crucial to translating our findings into targeted immunotherapeutic strategies.

4. Materials and Methods

4.1. Data Acquisition and Sources

This study is a analysis of previously published data. The single-cell RNA sequencing and ATAC sequencing data used in this study were primarily obtained from the Gene Expression Omnibus (GEO) database of the National Center for Biotechnology Information. The cerebrospinal fluid (CSF) dataset comprises a total of 77 10x Genomics 5' single-cell transcriptome samples (GSE200164 and GSE134577), including 24 Alzheimer's disease samples with a mean age of 69.5 years (SD 9.4) and 53 healthy control samples with a mean age of 69 years (SD 6.6). The blood dataset consists of 50 10x Genomics 5' single-cell transcriptome samples (GSE226602), including 25 Alzheimer's disease blood samples with a mean age of 72.5 years (SD 10.4) and 25 healthy control blood samples with a mean age of 72.9 years (SD 8.3) (Supplementary Tables S1 and S2). Additionally, 49 scATAC-seq datasets (GSE226267) matched with the blood scRNA-seq samples were collected to identify key genes and regulatory factors involved in immune infiltration. Microvascular single-cell transcriptome dataset (GSE252921) were included to study blood-brain barrier disruption. Parkinson's disease scRNA-seq data were downloaded from the provided link (<https://zenodo.org/record/3993994>, accessed on 5 December 2025) to investigate the prevalence of CD8⁺ TPex cells across diseases. The raw sequencing data for both modalities were processed using CellRanger (v1.1.0) to generate standard output files for scATAC-seq analysis.

4.2. scRNA-Seq Data Processing

The preprocessing and analysis of CSF and peripheral blood scRNA-seq data were conducted with the Seurat package (v5.2.1) [29]. Quality filtering retained cells with 200 to 6000 detected genes and a mitochondrial gene percentage below 10%. Normalization was performed using the LogNormalize method, followed by scaling to regress out covariates, including nCount_RNA, percent.mt, and cell cycle scores. Principal Component Analysis (PCA) was applied to variable features, and the first 15 principal components were selected for downstream analysis. We integrated datasets from different sources using the Harmony R package (v1.2.3) to perform batch effect correction and evaluated the integration effect through UMAP visualization. (Supplementary Figure S1). Cell clustering was performed on the integrated data using the FindNeighbors and FindClusters functions (Louvain algorithm, resolution = 0.6). Finally, potential doublets were detected and removed using the DoubletFinder package (v2.0.3) with an optimized pK parameter.

4.3. Cell Trajectory Inference

Cell trajectory inference was carried out via Monocle3 (v1.3.7) [11]. By integrating the single-cell transcriptome data of AD blood and CSF, the Seurat package was used for CCA dimensionality reduction and clustering in the same space, and the trajectory UMAP map was generated through the Monocle3 R package. In addition, we calculated the depletion score of CD8⁺ TPex cell types during the cell trajectory analysis process through AddModuleScore.

4.4. Cell–Cell Interaction Analysis

We inferred the interactions among CD8⁺ T cells, monocyte cells and other cells through CellChat (v1.6.1) [22]. Metacell expression profiles and metadata are used as the inputs of CellChat. CellChat uses the CellPhoneDB (receptor–ligand interaction database)

to model communication probabilities on the basis of the law of mass action and uses permutation tests to identify significant communications. The communication probabilities between all the inferred cell population pairs are represented by a three-dimensional array. CellChat analyzes and visualizes the inferred networks via social network metrics, pattern recognition methods, and manifold learning methods.

4.5. *scWGCNA*

scWGCNA performs weighted gene co-expression network analysis on single-cell transcriptome data and can be performed directly on cell-annotated Seurat objects. Through the hdWGCNA R package (v0.4.04) [12], 2000 highly variable genes were screened as the input genes of the Seurat objects. Reducing the influence of noise data on the subsequent analysis results and decreasing the feature dimensions of metacells can improve the analysis speed. The TestSoftPowers function determines the threshold required for the analysis. By analyzing the relationships between the modules and the levels of module characteristic genes (MEs), the modules enriched in the target cell type were selected. To analyze the key genes related to immune infiltration in the cell trajectory, the top 25 module genes according to the KME value of the cell type module according to hdWGCNA were identified. Then, LASSO regression and the RF algorithm were used to take the intersection, and the common genes were taken as the characteristic genes.

4.6. *scATAC-Seq Data Processing*

The scATAC fastq files obtained from the ENA database were processed into fragment files via cellranger-atac. The fragment file of each sample was loaded into R (v4.4.1) through the createArrowFiles function of the ArchR (v1.0.3) software package [30], and the gene activity score was calculated for each arrow file for downstream analysis. Next, quality control was carried out for each cell. Only cells with a TSS enrichment score > 4 and a number of unique fragments >2000 were retained. Dimensional reduction via 4 rounds of latent semantic indexing (LSI) was computed on the tile matrix via the addIterativeLSI function of ArchR. After that, the initial clustering was carried out through the addClusters function of the ArchR package, with a resolution of 0.3. Through the addGeneIntegrationMatrix function of ArchR, scATAC and scRNA were integrated for cell annotation. For each unsupervised clustering result, peak calling was performed via MACS2 (v2.2.9.1) [31].

4.7. *Differential Accessibility Analysis*

The ArchR object was converted into a Seurat object through the ArchR2Signac function of the signac R package (v1.14.0), and the difference peaks between cell types were analyzed through the FindMarkers function. The Wilcoxon rank sum test was used to test the differential peaks. When adjusted $p < 0.05$ and the absolute value of log2-fold change > 0.5, the peaks were identified as significantly different. The peak visualization of each gene was plotted via the CoveragePlot function of GenomicRanges (v1.60.0).

4.8. *ChromVar Deviation for TF Activity*

The motif files were obtained from the JASPAR2020 database, and scATAC was matched with the motif files through the AddMotifs function of Signac. Using the FindMotif function, the differential motifs of the obtained differential peaks were analyzed to retain the significant peaks with adjusted $p < 0.05$ and an absolute value of log2-fold change > 0.5.

4.9. TF Footprinting

The matchMotif function in the motifmatchr package was used to match the CIS-BP motif with the scATAC peak. After marking the specific TF motifs, the peaks with the specific TF motifs were extracted. The getFootprints function of ArchR was used for footprint analysis. To explain the Tn5 bias, K-mer ($k = 6$) sequences around each Tn5 insertion site were identified from each artifact and converted into a K-MER frequency table via the oligonucleotide frequency function in the Biostrings package. The genome-wide expected k-mers from the hg38 bsgenome were calculated via the same function. For each motif, the k-mer frequency matrix represents all possible k-mers within a 200 bp range upstream and downstream of the motif center and is generated by iterating at each motif site. A genome-wide k-mer frequency matrix was also generated via a similar method. The expected Tn5 insertion was estimated by multiplexing the k-mer position frequency table with the observed/expected Tn5 k-mer frequency. We subtracted the Tn5 bias from the footprint signal.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/ijms27041783/s1>.

Author Contributions: Y.Z., S.N. and H.Z. (Hui Zhi) conceived this study. M.X., Y.X., W.D., X.L., H.Z. (Hongbo Zhu), S.W., M.L., J.D., Y.L. and W.F. assisted with the implementation of the study and data analysis. Y.Z. wrote the paper. All authors have read and agreed to the published version of the manuscript.

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